

OECD Guidelines for the Testing of Chemicals (OECD TG) are a set of internationally accepted specifications for the testing of chemicals decided on by the Organisation for Economic Co-operation and Development (OECD). They were first published in 1981. They are split into five sections:

Section 1: Physical Chemical Properties

Section 2: Effects on Biotic Systems

Section 3: Environmental Fate and Behaviour

Section 4: Health Effects

Section 5: Other Test Guidelines

Guidelines are numbered with three digit numbers, the section number being the first number. Sometimes guidelines are suffixed with a letter.

Guidelines are under constant review, with guidelines being periodically updated, new guidelines being adopted, and guidelines being withdrawn. Previous guidelines are maintained on the website for reference purposes. Animal welfare concerns are dealt with by ensuring that animal tests are only permitted where necessary. An OECD Directive obligates a national regulator to use TG studies performed in another member country (Mutual Acceptance of Data, 'MAD' (enacting OECD's mission of reducing non-tariff trade barriers), and the OECD reaches out non-OECD countries to use the TG when those countries regulate chemicals.

Many of the TG for health effects (toxicity tests) function together, in a process called 'dose-ranging'; lowering doses while extending exposure period. The in vitro effects of a chemical are quickly discovered using a wide range of doses, which sets the doses for acute in vivo tests; next a semi-chronic exposure tests in vertebrates, whose purpose is to find the 'Maximally Tolerated Dose' (MTD)—the dose that the animals are likely to tolerate for the duration of the final TG—a chronic test of typically one to two years exposure (to mimic a

lifetime of human exposure). The MTD becomes the highest dose in the chronic exposure test, and typically one to three more lower dose levels are added (typically spanning about 20 to 100-fold), often in the 'mg/kg per day' range (e.g., 400, 150 & 50 mg/kg d-). See the TG 453 (Combined Chronic Toxicity/Carcinogenicity) for more on dose ranging.

At these high chronic exposure doses, toxicity is likely in some but not all animals, allowing a small, affordable number of animals. The risk assessment is finished by taking the TG's no- or the lowest-effect dose, dividing it by a safety factor (e.g. 100-fold) and comparing that 'safe dose' to the anticipated exposures.

It has been observed that these chronic TG doses are unrealistically high (close to the poisonous doses) for determining effects from our actual exposures—i.e. they do not actually test their hypothesis. Academic researchers in contrast are interested in the hypothesis, 'what effects from our actual exposures?', and to date have published at least 20,000 findings of chemicals' toxicity at low doses in vertebrate animals, at the rate of about 3 or 4 every day. The chronic TG have further insensitivities to find toxicity: they sacrifice the animals at the human equivalent of the early sixties of age—before most chronic diseases even manifest. Effects are detected only with visible light microscopes. Finally, they are performed by the party with a huge financial interest in the chemical being shown to be safe enough to market. Nevertheless, regulators, politicians and other stakeholders believe the TG are reliable to determine risks with.

The guidelines are available in both English and French.

List of guidelines

Section 1: Physical Chemical Properties

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References

External links

OECD Guidelines for the Testing of Chemicals